8.5 Dilations: Perimeter and Area

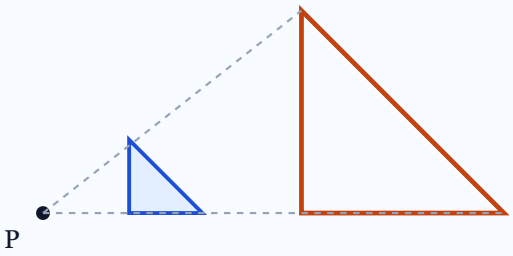
A dilation with scale factor k centered at the origin sends point (x, y) to (kx, ky) — every length in the figure is multiplied by $|k|$. Since perimeter is a sum of lengths, it scales the same way: $P' = |k| \cdot P$. Area, however, is a *product* of two lengths (like base times height), so it scales by k^2: $A' = k^2 \cdot A$. This squared relationship also works in reverse — if you know how areas compare, take a square root to find how sides or perimeters compare, and vice versa. This worksheet includes coordinate dilations and real-world scale-model problems that combine both relationships.

Examples

Example 1: Perimeter scales by k

A figure has perimeter 20. It is dilated with scale factor $k = 3$. Find the image's perimeter.

Solution



Every length in the image is k times the corresponding length in the original.

Identify the relationship: perimeter is a sum of side lengths, and every length scales by the factor k under a dilation, so $P' = k \cdot P$.
Compute: $P' = 3(20) = 60$.
Answer: 60.

Example 2: Area scales by k^2

A figure has area 12. It is dilated with scale factor $k = 2$. Find the image's area.

Solution

Identify the relationship: area is a product of two lengths (like base × height), and each of those lengths scales by k, so the area scales by k^2: $A' = k^2 \cdot A$.

Compute: $A' = 2^2(12) = 4(12) = 48$.

Answer: 48.

Example 3: Find k from a perimeter change

A perimeter maps from 18 to 45 under a dilation. Find k.

Solution

Identify: since perimeter scales linearly ($P' = k \cdot P$), the scale factor is the ratio of the new perimeter to the old one.

Compute: $k = \dfrac{P'}{P} = \dfrac{45}{18} = \dfrac{5}{2}$.

Answer: $k = \dfrac{5}{2}$ (or 2.5).

Example 4: Area with a fractional scale factor

A figure has area 40. It is dilated with $k = \dfrac{1}{2}$. Find the image's area.

Solution

Identify the rule: area scales by k^2, regardless of whether k is a whole number or a fraction.

Compute: $A' = \left(\dfrac{1}{2}\right)^2 (40) = \dfrac{1}{4}(40) = 10$.

Answer: 10. *Note a fractional $k < 1$ shrinks the figure, so the smaller area makes sense.*

Example 5: Side ratio to area ratio

Two similar figures have corresponding sides 6 and 9. Find the ratio of their areas (smaller to larger).

Solution

Identify the scale factor from the given sides: $k = \frac{9}{6} = \frac{3}{2}$.

Compute the area ratio by squaring the scale factor: $k^2 = (\frac{3}{2})^2 = \frac{9}{4}$.

Answer: $\frac{9}{4}$ (or $9:4$).

Example 6: Area ratio to perimeter ratio

Two similar figures have areas in ratio $4:1$. Find the ratio of their perimeters.

Solution

Perimeter (and side) ratio is the square root of the area ratio:
$\sqrt{4}:\sqrt{1} = 2:1$.
Answer: $2:1$.

Example 7: Reduction (shrinking)

A figure has perimeter 50. It is dilated with $k = 0.4$. Find the image's perimeter.

Solution

Identify: a scale factor less than 1 shrinks the figure, but the rule $P' = k \cdot P$ still applies directly.
Compute: $P' = 0.4(50) = 20$.
Answer: 20, which is smaller than the original, as expected since $k < 1$.

Example 8: Square — check both rules at once

A square has side 5. It is dilated with $k = 3$. Find the image's perimeter and area two ways each.

Solution

Perimeter: original $= 4(5) = 20$; using the scale rule, image $= 3(20) = 60$. Directly: new side $= 3(5) = 15$, so image perimeter $= 4(15) = 60$. ✓

Area: original $= 5^2 = 25$; using the scale rule, image $= 3^2(25) = 225$. Directly: new side $= 15$, so image area $= 15^2 = 225$. ✓

Example 9: Find the original area

After a dilation with $k = 3$, a figure's image has area 72. Find the original area.

Solution

Identify: this reverses the usual direction — solve $A' = k^2 A$ for the original area A by dividing instead of multiplying.

Compute: $A = \dfrac{A'}{k^2} = \dfrac{72}{3^2} = \dfrac{72}{9} = 8.$

Answer: 8.

Example 10: Coordinate dilation — verify both rules

A triangle has vertices $(0, 0)$, $(4, 0)$, $(0, 3)$. Dilate it from the origin with $k = 3$. Find the new vertices, and verify that the perimeter scaled by k and the area scaled by k^2.

Solution

New vertices: multiply each coordinate by 3: $(0, 0)$, $(12, 0)$, $(0, 9)$.

Original: legs 4, 3; hypotenuse $\sqrt{16 + 9} = 5$. Perimeter $= 4 + 3 + 5 = 12$. Area $= \dfrac{1}{2}(4)(3) = 6$.

Image: legs 12, 9; hypotenuse $\sqrt{144 + 81} = 15$. Perimeter $= 12 + 9 + 15 = 36$. Area $= \dfrac{1}{2}(12)(9) = 54$.

Check: $36 = 3(12)$ ✓ (perimeter $\times k$); $54 = 9(6) = 3^2(6)$ ✓ (area $\times k^2$).

Example 11: Application — scale model of a house

An architect's model house has a floor perimeter of 8 ft and a floor area of 3 ft^2. The real house is built at scale factor 50 (every model length corresponds to 50 times that length in real life). Find the real house's floor perimeter and floor area.

Solution

Perimeter: $P' = 50(8) = 400$ ft.

Area: $A' = 50^2(3) = 2500(3) = 7500$ ft^2.

Answer: real perimeter $= 400$ ft; real area $= 7500$ ft^2.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Perimeter

Perimeter 14; $k = 4$. Image perimeter?

Answer: **Verification:**

Problem 2: Area

Area 10; $k = 3$. Image area?

Answer: **Verification:**

Problem 3: Scale factor

Perimeter 24 maps to 8. Find k.

Answer: **Verification:**

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Problem 4: Area

Area 50; $k = \frac{1}{5}$. Image area?

Answer: **Verification:**

Problem 5: Area ratio

Sides in ratio $2:5$. Area ratio?

Answer: **Verification:**

Problem 6: Perimeter ratio

Areas in ratio $9:1$. Perimeter ratio?

Answer: **Verification:**

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Problem 7: Perimeter

Perimeter 36; $k = \frac{2}{3}$. Image?

Answer: **Verification:**

Problem 8: Area

Area 18; $k = 2$. Image?

Answer: **Verification:**

Problem 9: Find k

Areas 16 and 64 (original to image). Find k.

Answer: **Verification:**

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Problem 10: Side

Side 8; $k = 1.5$. Image side?

Answer: **Verification:**

Problem 11: Original area

Image area 45 after $k = 3$. Find the original area.

Answer: **Verification:**

Problem 12: Perimeter

Perimeter 100; $k = 0.25$. Image?

Answer: **Verification:**

Problem 13: Area ratio

Corresponding sides 4 and 6. Area ratio (first to second)?

Answer: **Verification:**

Problem 14: Challenge

A triangle with vertices $(0, 0)$, $(6, 0)$, $(0, 8)$ is dilated from the origin by $k = 1.5$. Find the perimeter and area of the image.

Answer: **Verification:**

Problem 15: Challenge

A poster's area increases from 24 in^2 to 96 in^2 under a dilation. If the original perimeter was 20 in, find the scale factor and the new perimeter.

Answer: **Verification:**

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✓ Answer Key

1. 56
2. 90
3. $\frac{1}{3}$
4. 2
5. 4 : 25
6. 3 : 1
7. 24
8. 72
9. 2
10. 12
11. 5
12. 25
13. 4 : 9
14. Perimeter = 36; area = 54
15. $k = 2$; new perimeter = 40 in

Great work — keep practising!